

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458390.7703 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.019
Transit depth (Rp/Rs)^2	3.0 +/- 0.65 [%]
Orbital Inclination (inc)	87.12 +/- 0.8
Airmass coefficient 1 (a1)	1.082 +/- 0.018
Airmass coefficient 2 (a2)	-0.072 +/- 0.075
Scatter in the residuals of the lightcurve fit is	1.68 %
Best Comparison Star	None
Optimal Aperture	4.79
Optimal Annulus	8.45
Transit Duration (day)	0.0833 +/- 0.0082

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.019 vs 0.1592 (deviation 0.73σ)
Duration fit vs published	0.0833 d vs 0.0946 d (deviation 1.38σ)
Mid-time O-C	-7.3 min
Depth SNR	9.1
Residual scatter	1.68 %

Light curve

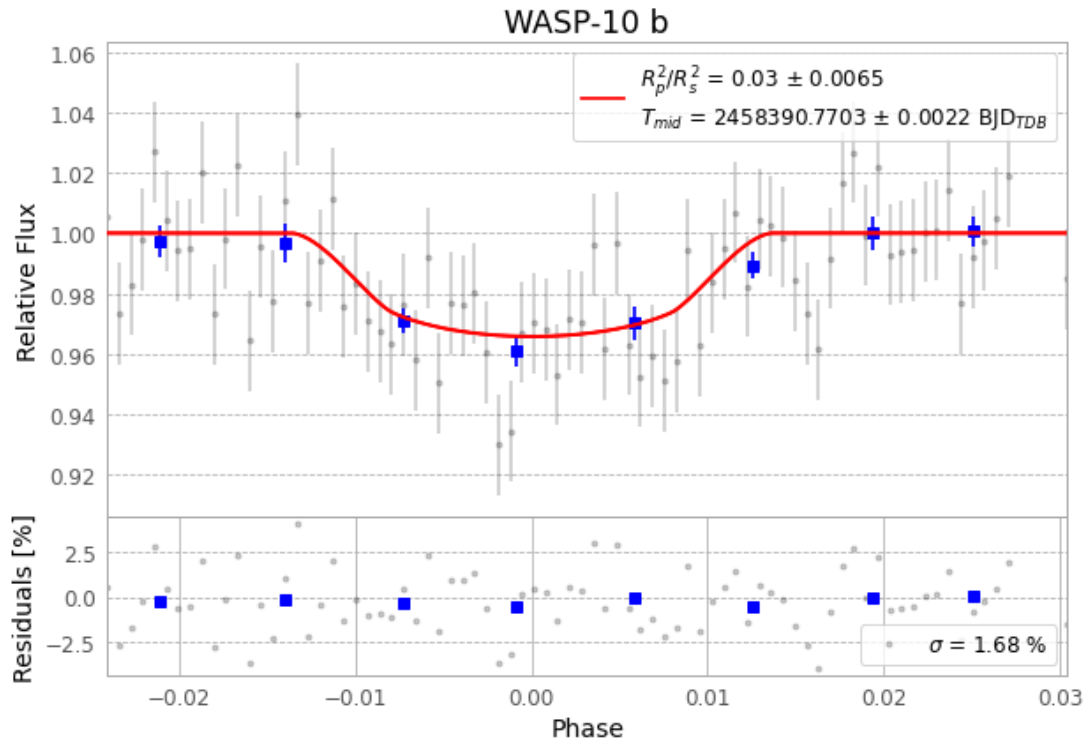


Figure 1 — FinalLightCurve_WASP-10 b_2018-09-28.png (SHA-256 979f801f6d371f3c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [344, 240], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f6884902e634eb0c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

83 FITS frames (55 MB), WASP-10180929043312.FITS → WASP-10180929083912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-180929.log (1e8bce2b53c0...), FinalLightCurve_WASP-10 b_2018-09-28.png (979f801f6d37...), FinalLightCurve_WASP-10 b_2018-09-28.csv (5a5db6114a03...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 17:34Z.